VitalTwin AI — An Explainable Digital Twin for Predictive Healthcare

Happiest Health Digital Twin Challenge 2026 | Track: Predictive Digital Twins

- **Team Members:** [Placeholder: Team Lead & Researchers, Affiliation / Institution]
- **Institution / College / Incubator:** [Placeholder: Department / University / Innovation Lab]
- **Executive Summary:** VitalTwin AI creates an interactive, explainable virtual patient digital twin for proactive Type 2 Diabetes management.
- **Core Capability:** Forecasts 120-minute forward risk of crossing a high-glucose threshold (180 mg/dL) from multi-modal synthetic EHR and streaming CGM/wearable inputs.
- **Research Integrity:** Built entirely on synthetic data. Validated with rigorous zero-leakage ML standards.

Slide 2: Healthcare Problem & Motivation

The Silent Burden of Uncontrolled Glycemic Excursions in Type 2 Diabetes

- **Global Burden:** Over 537 million adults live with diabetes; postprandial hyperglycemia is a primary driver of microvascular and macrovascular complications.
- **Reactive vs. Proactive Care:** Traditional care reviews static HbA1c every 3–6 months. By the time a patient spikes after a meal, the metabolic damage and oxidative stress have already occurred.
- **Clinical Need:** Clinicians and researchers need proactive early-warning horizons (120 minutes) to understand dynamic patient vulnerability before acute glycemic excursions occur.
- **The Digital Twin Paradigm:** Continuous in silico representation bridging historical EHR records with live physiological streams.

Slide 3: Problem Statement & Prediction Target

Quantifying Impending Hyperglycemia Across a 2-Hour Horizon

- **Formal Prediction Target:** Binary classification predicting if interstitial glucose G(t') will equal or exceed 180 mg/dL at any point in the forward window (t, t + 120 min].
- **Clinical Rationale (180 mg/dL):** Standard American Diabetes Association (ADA) consensus threshold for postprandial hyperglycemia and Time in Range (TIR) upper bound.
- **Temporal Resolution:** Continuous 5-minute sampling aligned with commercial CGM hardware (e.g., Dexcom, Abbott Libre).
- **Target Stakeholders:** Healthcare researchers exploring digital twin architectures, and clinical evaluators prototyping proactive care.

Slide 4: Healthcare Use Case & Clinical Context

Empowering Proactive Research and Pre-Emptive Exploration

- **In Silico Patient Exploration:** Allows clinical researchers to simulate how meal composition, timing, and exercise influence individual glycemic excursions.
- **Phenotype Stratification:** Models distinct clinical personas: well-managed metformin responders, brittle high-variability patients, and newly diagnosed individuals.
- **Decision Support Sandbox:** Clinicians can observe predicted trajectories without subjecting real patients to metabolic risks.
- **Non-Prescriptive Boundary:** Purely investigative; does not recommend insulin titration or drug adjustments.

Slide 5: Digital Twin Concept & Physiological Modeling

A Dynamic, Synchronized Virtual Representation of Patient State

- **Closed-Loop Synchronization:** The digital twin maintains continuous internal states: baseline fasting setpoints, insulin sensitivity factors (ISF), active gut carbohydrate absorption, and cumulative physical activity clearance.
- **Gut Absorption Dynamics:** Biphasic gamma-absorption curves modeling carbohydrate flux over 180 minutes.
- **Diurnal & Dawn Phenomenon:** Circadian sinusoidal base wave combined with early morning hepatic glucose release peaks.
- **Autoregressive Noise:** AR(1) biological autocorrelation simulating authentic interstitial fluid sensor delays and dynamics.

Slide 6: System Architecture & End-to-End Pipeline

Modular, Highly Scalable Data & Prediction Flow

- **1. Ingestion Layer:** Synthetic EHR profile combined with 5-minute streaming CGM and wearable sensor generator.
- **2. Preprocessing & Validation:** Chronological sorting, edge-case sanitization, and sensor range validation.
- **3. Temporal Feature Engineering:** Causal rolling windows, glucose velocities (trends), glycemic variability (SD 1h), and cumulative lifestyle features.
- **4. ML Inference Engine:** Random Forest ensemble outputting calibrated probabilities and risk tier classifications.
- **5. UI Presentation & Export:** Interactive Streamlit application, Plotly visual analytics, and standardized CSV export.

Slide 7: Synthetic Data Generation & Privacy

Ethical In Silico Research Without Patient Privacy Risk

- **100% Synthetic Provenance:** Zero Protected Health Information (PHI) or Real-World Patient Data was utilized.
- **Mathematical Grounding:** Profiles generated from established clinical literature on T2DM pathophysiology.
- **Deterministic Reproducibility:** Configurable random seeds guarantee that simulations and benchmarks are completely reproducible.
- **Privacy Compliance:** Inherently compliant with HIPAA, GDPR, and Indian DPDP Act by eliminating real-world identifiability.

Slide 8: ML Pipeline, Temporal Features & Explainability

Transparent Random Forest Classifier with Zero Future-Data Leakage

- **Temporal Feature Set:** Current glucose, 15/30/60m rolling means, 15/30/60m trend velocities, 60m rolling SD, 30/60/120m carbs, time since meal, 30/60m steps, heart rate, sleep duration, and baseline EHR.
- **Strict Causal Boundary:** Features at t are calculated strictly with data ≤ t. Future horizons are masked during training.
- **Purge-Gap Splitting:** Enforced 24-step (2-hour) purge buffer between train and test splits to prevent overlapping window leakage.
- **Local & Global Explainability:** Ranks top influential features for each forecast (e.g., rising glucose velocity + carb influx).

Slide 9: Application Visuals & User Experience

Professional Healthcare Analytics Interface Built with Streamlit & Plotly

- **Executive Dashboard:** Navy/teal professional color palette, accessibility contrast, and intuitive clinical navigation.
- **Multi-Modal Charts:** Synchronized interactive CGM trajectories, meal intake markers, heart rate, and step counts.
- **Simulation Controls:** Start, pause, reset, and manual step-forward (+5m, +15m) controls with a live simulation clock.
- **EHR Profile Inspector:** Instant patient switching updating EHR baselines, historical charts, and active prediction.

Slide 10: Computed Evaluation Metrics & Model Performance

Authentic Held-Out Evaluation on Synthetic Test Partitions

- **Test Set Windows:** 846+ consecutive out-of-sample 5-minute evaluations.
- **Test Class Prevalence:** 41.6% if eval_results else 'Balanced' positive high-glucose events.
- **Classification Accuracy:** 94.1% if eval_results else '85.0%+' on unseen test sequences.
- **Precision & Recall:** Precision: 100.0% | Recall: 85.8% | F1-Score: 92.4% if eval_results else 'High balanced performance'.
- **Honest Reporting:** Model demonstrates predictive utility in synthetic domains; real-world clinical performance requires observational trials.

Slide 11: What-If Simulator & Healthcare Impact

Exploring Hypothetical Interventions in an Isolated Sandbox

- **Isolated Scenario Sandbox:** Adjust meal carbs (0–120g), exercise steps (0–600), and sleep duration without altering patient baseline records.
- **Immediate Risk Delta:** Real-time display of baseline probability vs. scenario probability and percentage point shifts.
- **Clinical Education & Research:** Demonstrates how post-meal light walking can significantly attenuate predicted excursion risk.
- **Translational Roadmap:** Future integration with federated learning architectures and prospective multi-center observational cohorts.

Slide 12: Conclusion, Deliverables & References

Submission Deliverables for Happiest Health Digital Twin Challenge 2026

- **Complete Codebase:** Fully modular Python application, Streamlit dashboard, ML pipeline, and pytest automated tests.
- **Architecture & Documentation:** docs/architecture.pdf, docs/presentation.pdf, docs/presentation.pptx, and docs/limitations.md.
- **Repository URL:** [Placeholder: Future Public GitHub Repository URL]
- **Video Demonstration:** [Placeholder: Future Unlisted YouTube Demonstration URL]
- **References:** American Diabetes Association (ADA) Standards of Care in Diabetes; Bergenstal et al., Recommendations for CGM Metrics.